AI AND IOT-ENABLED SMART OFFICE SYSTEM WITH CLOUD-BASED ENERGY OPTIMIZATION

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DECLARATION

I, M.A.H.S.THATHSARANI (IT22564672), hereby declare that this thesis, titled "AI and IoT-Enabled Smart Office System with Cloud-Based Energy Optimization," is my own original work. It has been written and completed independently, except where specific reference has been made to the work of others. This thesis has not been submitted, either in whole or in part, to any other university or institution in partial fulfilment of any degree, diploma, certificate, or other qualification.

I confirm that all sources of information used in this work have been appropriately acknowledged in the References section, in accordance with IEEE citation standards.

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SUPERVISOR'S APPROVAL

This thesis titled "AI and IoT-Enabled Smart Office System with Cloud-Based Energy Optimization" has been reviewed and approved as a partial fulfillment of the requirements for the degree of ____________. The work presented is considered satisfactory in terms of scope, quality, and academic standards.

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ABSTRACT

The increasing demand for intelligent workplace environments has led to the integration of Artificial Intelligence (AI) and real-time monitoring systems to improve productivity and employee well-being. This research presents the design and development of a Smart Office Intelligence Platform that utilizes machine learning techniques to detect employee stress levels, dynamically assign tasks, and provide real-time system monitoring through an interactive dashboard. The proposed system focuses on three core components: stress detection, workload-based task allocation, and a smart dashboard interface.

The stress detection component employs a camera-based approach to capture employee facial data, which is processed using a machine learning model to classify stress levels into three categories: low, normal, and high. This enables the system to monitor employee conditions continuously and provide valuable insights into workplace well-being. The task assignment component utilizes the detected stress levels to dynamically allocate tasks based on workload classification. Tasks are categorized into low, medium, and high workloads, and an intelligent decision mechanism ensures that employees with higher stress levels are assigned lighter tasks, thereby maintaining productivity while reducing burnout. The smart dashboard component integrates these functionalities into a unified platform, enabling administrators and employees to interact with the system in real time. The dashboard provides features such as task management, stress monitoring, and camera request handling using modern web technologies and WebSocket-based real-time communication.

The system is implemented using a full-stack architecture comprising FastAPI for backend services, Next.js for frontend development, and SQLite for data management. A placeholder machine learning model is currently used for stress classification, achieving approximately 90% simulated accuracy, with future integration planned using deep learning techniques such as Convolutional Neural Networks (CNNs). Real-time communication between system components is

achieved using WebSockets, ensuring minimal latency in data updates and user interactions.

Experimental testing demonstrates that the system successfully integrates stress detection, task allocation, and real-time monitoring into a cohesive platform. The implementation highlights the potential of AI-driven solutions in improving workplace efficiency, optimizing task distribution, and enhancing employee well-being. The proposed system provides a scalable foundation for future enhancements, including advanced machine learning models, cloud deployment, and integration with additional IoT devices for comprehensive smart office automation.

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TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ABBREVIATIONS

Table 0-1. List of Abbreviations

Abbreviation	**Full Form**
AI	Artificial Intelligence
JSON	JavaScript Object Notation
LightGBM	Light Gradient Boosting Machine
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
MQ2	Metal Oxide Semiconductor Gas Sensor (model 2)
pkl	Pickle File (serialized Python object)
QoS	Quality of Service
RF	Random Forest
RMSE	Root Mean Square Error
REST	Representational State Transfer
SME	Small and Medium Enterprise
SVM	Support Vector Machine
TOC	Table of Contents
USB	Universal Serial Bus
XGBoost	Extreme Gradient Boosting

LIST OF APPENDICES

Chapter 1: INTRODUCTION

1.1.Introduction

Smart workplaces have been increasingly popular in recent years, thanks to the rapid growth of Artificial Intelligence (AI), Machine Learning (ML) and real-time monitoring technologies. Conventional workplaces are often managed manually without taking into account the well-being of employees, workload distribution, or real-time monitoring. This leads to problems like stress, mismanaged task allocation and low productivity [1], [2].

Stress in the workplace is a key factor affecting employee performance, satisfaction and productivity. Research in affective computing and human-centered AI has demonstrated that emotional and stress levels can be assessed by analysing facial expressions and physiological signals [6], [17]. But current office management systems do not have the ability to dynamically track stress levels and allocate tasks based on employee stress states. Moreover, in many workplaces, task assignment is not carried out in a condition-dependent manner, resulting in over- or under-estimation of human resources [25], [26].

Recent advances in computer vision and deep learning have made it possible to create a facial emotion recognition system with high accuracy to detect stress-related facial expressions from images and video feeds [1], [3]. This makes it possible to build smart systems which can track employee status in real time. In addition, machine learning algorithms have been extensively used in workload classification and decision making, enabling systems to allocate tasks according to various factors, including stress levels, past performance and task difficulty [23], [27].

This project aims to provide a solution to these issues through a Smart Office Intelligence Platform that combines machine learning and real-time communication techniques to build a smart office. The platform aims to track employee stress levels through a camera-based solution, classify the stress into a number of pre-defined categories and allocate tasks according to workload suitability. Furthermore, the system is equipped with a smart dashboard that enables real-time interaction between the administrators and employees via WebSocket communication.

The system has three components. The first component is the employee stress level classification using a machine learning approach. The second component is an intelligent task allocation system based on workload classification and employee status. The last component is the smart dashboard, offering real-time monitoring, task management and communication.

The application is developed with contemporary technologies such as FastAPI for the backend, Next.js for the frontend, and SQLite for database management. This is facilitated by using WebSockets for real-time communication, allowing for timely updates and interactions. The system employs a dummy machine learning model for stress detection, with the potential to integrate more sophisticated deep learning models, including Convolutional Neural Networks (CNNs) [1], [10] in the future.

This project has the potential to boost productivity, promote worker well-being, and showcase the benefits of AI-based systems in the workplace. The use of stress monitoring, smart task management, and real-time monitoring showcases the role of smart systems in creating responsive and productive workplaces [28], [30].

1.2. Background Literature

The development of Artificial Intelligence (AI), Machine Learning (ML) and Internet of Things (IoT) has significantly affected smart office systems. Recent research indicates the importance of including smart monitoring systems to enhance productivity, resource allocation and employee welfare. The rigidity of the traditional office systems and the need to adapt with real-time status of the employees have driven researchers to explore office optimisation using AI [1], [28].

A major area of research relevant to this project is facial emotion recognition (FER) which is vital for emotion stress detection. Deep learning based Convolutional Neural Networks (CNNs) have achieved remarkable results in facial emotion recognition from still images and video sequences [1], [10]. Large databases such as AffectNet and DEAP have also helped to build robust emotion recognition systems by providing labelled facial and physiological data [2], [17]. This has enabled the real-time detection of stress to integrate with smart systems.

Besides the analysis of facial expressions, some studies have reported multimodal approaches for stress detection, which include the combination of facial data with physiological data (e.g. heart rate, skin conductance and EEG) [5], [16]. While these methods are more precise, they may require extra equipment, which may not be practical in the workplace. Consequently, video systems have become more prevalent due to their affordability, convenience and integration with other systems.

An important research focus is workload classification, where workload is measured using machine learning techniques. Workload classification has been shown to be feasible using data such as behaviour patterns, physiological data and computer activities [23]. This is vital in the development of intelligent systems that can automatically assign tasks based on the individual's capacities and efficiency.

Task allocation and optimisation has also been extensively studied in distributed systems and scheduling. Random Forests and decision trees have been used to predict task allocation based on various input variables such as skill, workload and system constraints [25], [27]. This increases efficiency by allocating tasks smartly and effectively.

Also, real-time monitoring and smart dashboards have been extensively studied in smart factories and offices. Existing dashboards are built using data visualization, real-time communication and decision-making, to provide timely insights to users [29]. Using WebSocket communication between system components provides real-time updates, making it possible to see changes in the system in real-time and improve responsiveness.

But most of the existing systems are either focused on stress detection, task allocation or monitoring, but not both. This suggests the need for a system that provides stress detection, smart task classification and real-time monitoring to create a smart office system.

1.2.1. Significance of Study

The current research is significant because it helps develop a smart office management system that enhances productivity, comfort and task efficiency. Stress is an important element affecting productivity, performance and job satisfaction in today's offices. Current office management systems lack mechanisms of monitoring employees' stress and scheduling tasks accordingly, leading to poor task allocation and even burnout [6, 21].

In this paper, we propose a Smart Office Intelligence system with machine learning and real-time monitoring. The system is designed to include a camera-based stress monitoring system to track employees' stress by analysing their facial expressions. Recent advances in facial emotions recognition and deep learning techniques has demonstrated the viability of such approach for identifying emotions and stress in real-time applications [1], [3]. This presents an efficient and practical way to implement stress-aware systems without complex hardware setups.

Further, the integration of stress detection and task allocation algorithms offers added value in the workplace. The task allocation method, which is based on employees' stress levels, helps achieve a fair distribution of workload, which results in better productivity and reduced stress and fatigue problems. The task allocation mechanisms based on machine learning have been applied successfully in other resource allocation and decision-making problems [25], [27].

A further contribution of this study is the development of a smart dashboard for the interaction between administrators and the employees in real-time. Using WebSocket, the system will stream updates to the user which will allow administrators monitor employees as well as other system events in real time. Real-time monitoring systems are well recognised for improving responsiveness and decision making in real-time environments [29].

Overall, the research is significant as it demonstrates how artificial intelligence, real-time communication, and modern web technologies can be used to develop an adaptive smart office. The system not only improves task allocation and efficiency, but also employee quality of life by taking into account psychological factors such as stress. This research provides the foundation for smart office systems with advanced machine learning algorithms, IoT devices and scalability to be used by large organisations [28], [30].

1.2.2. Research Scope

This study aims to explore the development of a Smart Office Intelligence Platform that leverages machine learning, real-time communication and advanced web technologies to enhance productivity and well-being in the workplace. The research mainly focuses on three major aspects of the functionality: stress detection; workload-based intelligent task allocation; and system monitoring via a smart dashboard.

The first component of the research scope relates to the development of a stress detection system using a camera. The system monitors employees' faces using an ordinary web camera, analyzing them through machine learning methods to categorise their stress levels into levels such as low, medium and high stress. Recent studies have extensively investigated the use of facial emotion recognition techniques, which have been successful in detecting emotional and stress states using computer vision and deep learning techniques [1], [3]. In this paper, a basic model is developed to mimic the stress classification process, with the aim of being able to incorporate more sophisticated deep learning models in the future.

The second part of the study involves smart task allocation using workload classification. It classifies tasks according to workload and dynamically allocates tasks based on the stress level of employees. This helps to balance the workload and avoid stressful or under-productive situations. Task allocation and decision-making using machine learning algorithms have been extensively used in project management and distributed systems to enhance resource management and efficiency [25], [27]. In this work, a rule-based system is employed with the option of incorporating predictive machine learning models in the future.

The third part of the research focuses on the design of a smart dashboard for real-time communication between managers and employees. The real-time dashboard offers features like employee presence monitoring, camera requests, task management, and task status updates. Bi-directional real-time data streams are created using WebSocket, which also allows for low-latency communication and improves responsiveness. The need for real-time data presentation and interaction is highlighted in smart monitoring systems for decision-making and control [29].

Besides these essential elements, the system also includes other features, including user authentication using JWT, password hashing using bcrypt, and database storage using SQLite. The system uses FastAPI for the backend and Next.js and Tailwind CSS for the frontend to ensure a responsive and interactive user interface.

But this study is confined to a development environment and does not involve deployment at a large scale or real-time connection with external IoT devices. The stress detection system is based on a mock machine learning model without a trained deep learning model. In addition, this research does not explore the effects of environmental changes, different facial expressions and workplace scenarios.

Despite these constraints, the proposed system offers a holistic approach to incorporate stress detection, task management and real-time monitoring in a smart office system. This research sets the groundwork for future work, which can include the implementation of more advanced machine learning models, cloud deployment, and extension to fully automated workplace environments [28], [30].

1.2.3. Target Audience

The proposed Smart Office Intelligence Platform is designed to benefit a wide range of users and stakeholders involved in modern workplace environments. The primary target audience includes organizations and businesses that aim to improve productivity, employee well-being, and operational efficiency through the adoption of intelligent technologies.

One of the key target groups is organizational management and administrators, who are responsible for overseeing employee performance and task allocation. The system provides administrators with real-time insights into employee stress levels, task progress, and overall system activity through an interactive dashboard. This enables informed decision-making and efficient resource management, which are essential for maintaining productivity in dynamic work environments [29].

Another important target audience is employees working in office environments, particularly those engaged in knowledge-based or high-pressure tasks. By monitoring stress levels and dynamically adjusting workload, the system helps reduce mental fatigue and prevent burnout. Research has shown that employee stress significantly impacts job performance and overall organizational outcomes, highlighting the importance of systems that support employee well-being [6], [21].

The system is also relevant to human resource (HR) professionals, who can utilize the platform to monitor employee conditions, analyze workload distribution, and implement strategies to improve workplace satisfaction. By integrating machine learning-based stress detection and workload classification, the system provides valuable data that can support employee management and performance evaluation processes [25], [27].

In addition, the platform can be beneficial for researchers and developers working in the fields of artificial intelligence, machine learning, and smart systems. The integration of stress detection, task allocation, and real-time dashboards provides a practical framework for further research and development in intelligent workplace solutions. Studies in affective computing and smart monitoring systems emphasize the growing importance of such integrated approaches in modern technological applications [1], [28].

Furthermore, the system may also be applicable to industrial environments and smart organizations that are transitioning toward Industry 4.0 and digital transformation. The use of real-time monitoring, intelligent decision-making, and automated task management aligns with the principles of smart systems and cyber-physical architectures [28], [30].

Overall, the target audience of this research includes administrators, employees, HR professionals, researchers, and organizations seeking to adopt intelligent systems for improving workplace efficiency and employee well-being.

1.2.4. Research Questions

This study is focused on answering a number of important questions based on the problem identified and the research scope of the study. The research questions are posed to inform the Smart Office Intelligence Platform's design, development, and testing.

The main research question of this research is:

How can machine learning and real-time monitoring technologies be combined to create an intelligent system that boosts employee productivity with the right stress level?

The sub-research questions that address this primary question are:

How can we detect the stress of employees via a computer vision system?

Research in facial emotion recognition and affective computing has shown the feasibility of using computer vision techniques to infer emotions, but there are still challenges in the real-time detection of stress in workplace scenarios [1], [3].

How can stress level classification be incorporated to improve task allocation in workplaces?

While task assignment using machine learning has been applied to enhance productivity, incorporating human considerations such as stress into decision-making is still being explored [25], [27].

How can tasks be classified into workload categories (low, normal, high) to achieve task balance?

The classification of workload is important to avoid burnout and increase productivity, as discussed in research on the assessment and management of mental workload [23].

How can real-time communication systems (WebSockets) be incorporated to improve communication between the administrator and employees?

Real-time systems are crucial for dynamic system environments, providing immediate updates and facilitating quicker decision-making [29].

How can a unified dashboard system enhance monitoring, control and user friendliness of a smart office platform?

Intelligent dashboards have been proven useful in providing insights and enhancing efficiency in smart systems and environments [28], [30].

These questions cover the key elements of the system, such as stress detection, task scheduling, workload classification and real-time system interaction. Through addressing these questions, the research will show the viability and benefits of combining machine learning and real-time systems in building a smart, adaptive work environment.

1.3. Research Gap

To understand the limitations of existing systems and justify the proposed Smart Office Intelligence Platform, a comparative analysis was conducted against recent research studies. The table below highlights key gaps in current approaches and demonstrates how the proposed system addresses these limitations.

With the rapid advancement of artificial intelligence and smart workplace technologies, several research studies have focused on areas such as stress detection, task allocation, and employee monitoring. However, these studies are often limited to isolated functionalities and lack a comprehensive, integrated approach.

Existing research on stress detection using facial expressions and machine learning models [1], [5], [11] primarily focuses on identifying emotional states but does not utilize this information for decision-making processes such as task assignment. Similarly, machine learning-based task allocation systems [25], [27] aim to optimize resource utilization and productivity, yet they do not consider the psychological condition or stress level of employees during task assignment.

Moreover, monitoring systems and smart dashboards [29] provide real-time visualization of employee activities but lack intelligent automation and adaptive capabilities. These systems do not integrate stress detection or predictive workload classification, resulting in limited effectiveness in improving employee well-being and performance.

Another significant limitation is the absence of real-time interaction mechanisms, such as WebSocket-based communication, which are essential for enabling dynamic updates and instant responses within a smart office environment. Additionally, most existing systems do not support features like camera-based monitoring requests or automated task reallocation based on employee conditions.

Table 1.1 presents a comparative gap analysis, contrasting the proposed system against five representative existing studies. The analysis highlights that no prior work has simultaneously combined all capability dimensions that define the proposed system.

Table 1-1. Research Gap Analysis – Proposed System vs. Existing Studies

Feature / Capability	Proposed System	Li & Deng [1]	Hosseini et al. [5]	Bhattacharyya et al. [11]	Al-Tarawneh et al. [25]	Woo et al. [29]
Real-Time Stress Detection (Camera-Based)	Yes	No	Partial	Yes	No	Partial
Stress Classification (Low / Normal / High)	Yes	Yes	Partial	Yes	No	No
ML-Based Stress Prediction	Yes	Yes	Yes	Yes	No	Partial
Task Assignment Based on Stress Level	Yes	No	No	No	Yes (no stress)	No
Task Workload Classification (ML)	Yes	No	No	No	Yes	Partial
Dynamic Task Reallocation	Yes	No	No	No	Partial	No

Feature / Capability	Proposed System	Li & Deng [1]	Hosseini et al. [5]	Bhattacharyya et al. [11]	Al-Tarawneh et al. [25]	Woo et al. [29]
Real-Time Communication (WebSocket)	Yes	No	No	No	No	Partial
Smart Dashboard (Admin + Employee)	Yes	No	No	No	No	Yes
Employee Monitoring System	Yes	No	Partial	Yes	No	Yes
Integrated System (ML + Tasks + Dashboard)	Yes	No	No	No	No	Partial
Camera Request Feature (Admin → Employee)	Yes	No	No	No	No	No
Full Automation (Stress → Task → Monitoring)	Yes	No	No	No	No	No

The comparison reveals several important limitations in existing studies:

- Most systems focus only on one domain, such as anomaly detection, network monitoring, or emotion recognition, rather than providing a fully integrated solution.
- Existing research lacks real-time adaptive task allocation based on employee stress levels, which is a key feature of the proposed system.
- While some studies implement machine learning models, they do not combine:
    - Stress detection
    - Task classification
    - Real-time monitoring

    into a single platform.
- Real-time communication mechanisms such as WebSockets are rarely used, limiting system responsiveness.
- Many systems do not provide an interactive dashboard for administrators to monitor and control operations effectively.

1.4. Research Problem

Modern workplaces are becoming increasingly complex, requiring efficient task management and continuous monitoring of employee performance. However, existing systems fail to address the dynamic relationship between employee mental states and productivity. Traditional task allocation methods assign work based on static factors such as availability or predefined roles, without considering the employee's real-time stress level or cognitive workload.

Recent advancements in machine learning and computer vision have enabled stress detection through facial expressions and physiological signals [1], [5], [11]. However, these approaches are typically limited to analysis and do not extend to actionable decision-making processes such as task assignment or workload balancing. Similarly, task allocation systems based on machine learning techniques [25], [27] focus on optimizing productivity but ignore the human factor, particularly employee stress and well-being.

Furthermore, existing workplace monitoring systems [29] provide dashboards for tracking employee activities but lack integration with intelligent models that can adapt task assignments dynamically. This results in inefficiencies such as overloading employees under stress, reduced productivity, and potential burnout.

Therefore, the main research problem can be stated as:How to design and develop an intelligent smart office system that integrates real-time stress detection, machine learning-based task allocation, and interactive monitoring dashboards to improve employee well-being and organizational productivity.

1.5. Sub-Problems

To effectively address the main research problem of developing an intelligent Smart Office system, it is necessary to break it down into smaller, manageable sub-problems. These sub-problems focus on the core functionalities of the system and guide the development of each component.

1.6. Identified Sub-Problems

1.6.1. Real-Time Stress Detection

- How to accurately detect employee stress levels using facial expressions in real-time.
- How to ensure reliability under different lighting conditions and facial variations.
- How to classify detected stress into categories such as Low, Normal, and High using machine learning models.

1.6.2. Machine Learning-Based Stress Classification

- How to train and optimize a machine learning model for accurate stress prediction.
- How to preprocess image data and extract relevant features for classification.
- How to ensure the model performs efficiently in real-time environments.

1.6.3. Task Workload Classification

- How to categorize tasks based on workload levels (Low, Normal, High).
- How to apply machine learning techniques to predict task difficulty.
- How to maintain a balanced workload distribution among employees.

1.6.4. Intelligent Task Assignment

- How to dynamically assign tasks to employees based on their current stress level.
- How to prevent overloading employees with high stress levels.
- How to implement automatic task reallocation when stress levels change.

1.6.5. Real-Time Communication System

- How to establish real-time communication between admin and employees using WebSockets.
- How to send instant notifications such as camera requests or task updates.
- How to ensure reliable and low-latency communication.

1.6.6. Smart Dashboard Development

- How to design an interactive and user-friendly dashboard for administrators and employees.
- How to visualize real-time data such as stress levels, tasks, and attendance.
- How to provide control features such as task creation and monitoring.

1.6.7. System Integration

- How to integrate all components (ML models, backend, frontend, database) into a unified system.
- How to ensure smooth data flow between modules.
- How to maintain system scalability and performance.

1.7. Research Aim and Objectives

1.7.1 Aim

The aim of this research is to design and develop an intelligent Smart Office Intelligence Platform that integrates machine learning, real-time monitoring, and modern web technologies to improve workplace productivity and employee well-being by detecting stress levels, classifying workloads, and dynamically assigning tasks.

1.7.2 Main Objective

The main objective of this study is to develop a fully integrated system that:

- Detects employee stress levels in real time using a camera-based machine learning approach
- Classifies tasks based on workload levels (Low, Normal, High)
- Dynamically assigns tasks to employees based on their stress conditions
- Provides a smart dashboard for real-time monitoring and interaction

1.7.3 Sub Objectives

To achieve the main objective, the following sub-objectives are defined:

Stress Detection and Classification

- To capture employee facial data using a webcam
- To develop a machine learning model that classifies stress levels into Low, Normal, and High
- To ensure real-time processing and accurate prediction

Task Workload Classification

- To categorize tasks based on workload levels
- To apply machine learning techniques for workload classification
- To maintain balanced workload distribution among employees

Intelligent Task Assignment

- To dynamically assign tasks based on employee stress levels
- To implement a rule-based or ML-based decision system for task allocation
- To enable automatic task rotation after task completion

Smart Dashboard Development

- To design an interactive dashboard for administrators and employees
- To display real-time information such as stress levels, task assignments, and attendance
- To provide features for task management and monitoring

Real-Time Communication

- To implement WebSocket-based communication for instant updates
- To enable features such as camera requests and notifications
- To ensure low-latency and reliable system interaction

System Integration

- To integrate frontend, backend, database, and machine learning components
- To ensure smooth data flow across all modules
- To maintain system performance and scalability

1.8. Organization of the Chapters.

This report is organized into four main chapters, each addressing a specific aspect of the Smart Office Intelligence Platform, from introduction to final conclusions.

1.8.1. Chapter 1: Introduction

The research investigation is thoroughly summarized in this chapter. The background literature, the importance of the study, the research scope, the target audience, the research questions, the research gap, the research challenge, and the research objectives are all included. This chapter lays forth the rationale and framework for the suggested system.

1.8.2. Chapter 2: Methodology

The suggested system's design and development methodology is covered in this chapter. It describes the components, procedures, and system architecture related to job delegation, dashboard implementation, and stress detection. It also covers the integration of all system components, hardware and software technologies, and machine learning models.

1.8.3. Chapter 3: Results and Discussion

The results of the system's implementation and testing are presented in this chapter. The dashboard functionality, work assignment logic, and stress detection model performance are all assessed. Additionally, the chapter discusses, analyzes, and observes how the system behaves in various settings.

1.8.4. Chapter 4: Conclusion

This chapter provides an overview of the Smart Office Intelligence Platform's contributions and overall study findings. It emphasizes how effectively the suggested approach works to increase worker well-being and workplace productivity. It also addresses its shortcomings and makes recommendations for future developments.

1.9. Chapter Summary

The rationale and significance of creating intelligent workplace systems were highlighted in this chapter, which also provided an overview of the Smart Office Intelligence Platform. The underlying literature on stress detection, workload categorization, and smart monitoring systems was covered, with a focus on the necessity of combining these technologies into a cohesive solution.

The study's importance was discussed, showing how the suggested strategy enhances worker satisfaction and company output. After defining the research scope and target audience, the study's guiding research questions were developed. After a thorough examination of the research gap revealed the shortcomings of current systems, the primary research challenge and its related subproblems were defined.

In addition, the goals and purpose of the study were well-defined in order to tackle the problems that were found. In order to give readers an organized comprehension of the report as a whole, the chapter arrangement was finally offered.

Overall, by identifying the issue area and describing the proposed Smart Office Intelligence Platform's orientation, this chapter lays a solid basis for the investigation.

Chapter 2: METHODOLOGY

2.1. Introduction to Methodology

This chapter outlines the design and development process used for the Smart Office Intelligence Platform. The methodology describes the methodical process taken to execute the system's essential elements, such as job delegation, stress detection, and the creation of smart dashboards. The system architecture, data flow, machine learning methods, and technologies utilized to accomplish the study goals are explained in depth.

The suggested solution uses a full-stack development methodology, combining database, machine learning, frontend, and backend components into a single framework. In order to detect stress, the process starts with data collecting, which involves employing a camera to record employee face data. After that, a machine learning algorithm uses this data to categorize stress levels into low, normal, and high groups. The job assignment module receives the categorized stress level as input.

Employee stress levels are mapped to suitable workload categories using a decision-making mechanism in the job assignment component. The system automatically distributes tasks to employees based on their present state, and assignments are categorized into low, normal, and high workload levels. This guarantees a fair allocation of workloads and lessens performance problems brought on by stress.

The system also has a smart dashboard that allows administrators and staff to monitor and communicate in real time. In order to offer real-time updates, notifications, and system management functions, the dashboard incorporates WebSocket-based communication and is built using contemporary web technologies.

Additionally, the approach places a strong emphasis on system integration, in which real-time communication protocols and RESTful APIs connect every component. Tailwind CSS and Next.js are used to create the frontend, while FastAPI is used to construct the backend. System data, such as tasks, stress levels, and user information, is managed using a relational database.

All things considered, this chapter offers a thorough explanation of the techniques and tools utilized to create the suggested system, guaranteeing that all study goals are successfully reached using a methodical and scalable strategy.

2.2. Methodology

2.2.1. Dataset Description

Two distinct datasets underpin the six ML models in this system. Table 2.1 provides a summary of the key characteristics of each dataset.

Table 2-1.Dataset Summary – Six ML Models

Model	Dataset	Dataset Size	Features	Labels / Target	Imbalance
Stress Detection ML Model	AffectNet / FER Dataset	~400,000 images (AffectNet) / ~35,000 (FER2013)	Facial images, pixel intensity values, facial landmarks	Low Stress, Normal Stress, High Stress (mapped from emotions)	Yes (more neutral & happy samples)

Task Scheduling ML Model	Custom Task Dataset / Simulated Data	Structured dataset (tasks + employee data)	Stress level, task type, task duration, workload level	Low Workload, Normal Workload, High Workload	Possible imbalance depending on task distribution

The system utilizes two main machine learning models to achieve intelligent decision-making.

AffectNet and FER2013 are two facial emotion recognition datasets that serve as the foundation for the Stress Detection Model. These datasets include captioned photos of faces that depict various emotional states. These feelings are translated into three stress levels for this system: Low, Normal, and High. Because of its efficiency in image classification tasks, a Convolutional Neural Network (CNN) is employed.

The job Scheduling Model makes use of structured data that includes workload details, job characteristics, and employee stress levels. This methodology allocates jobs based on their classification into workload categories. Because of its stability, interpretability, and effective handling of structured data, a Random Forest classifier is chosen.

2.2.2. Methodological Structure

The suggested Smart Office Intelligence Platform's methodological structure is intended to combine various system elements into a cohesive framework that facilitates real-time communication and intelligent decision-making. Data collecting, processing, machine learning, decision-making, and display are among the interrelated layers that make up the system's structure.

Data collection is the methodology's initial step, during which input data is gathered from various sources. The employee's webcam is used to take face pictures for the stress detection component. Structured data, including job specifics, workload levels, and employee status, are gathered and kept in the database for the task scheduling component.

Data preparation is the second step, when the gathered information is made ready for analysis. To guarantee accuracy and consistency, image data is resized, normalized, and face detected. In a similar manner, structured task data is prepared for the machine learning model by organizing and formatting it.

Two models make up the machine learning processing layer, which is the third level. The first model is the stress detection model, which divides stress levels into low, normal, and high categories based on facial image analysis. The second model is the task scheduling model, which categorizes and assigns suitable workload levels based on job-related characteristics and employee stress levels. These models comprise the system's fundamental intelligence.

The decision-making layer, which comes after the machine learning stage, uses the model outputs to decide how the system should operate. The system chooses appropriate tasks and allocates them to staff based on the anticipated stress level. This keeps workers from being overworked under stressful circumstances and guarantees a fair job allocation.

The communication layer, which comes next, allows system components to communicate with one another in real time. Instantaneous updates, including camera requests, job assignments, and status updates, are made possible via WebSocket technology. This guarantees communication with minimal latency and improves system responsiveness.

The smart dashboard interface, which enables administrators and staff to engage with the system, is the last component of the presentation layer. Stress levels, work assignments, and system alerts are all shown in real-time on the dashboard.

Additionally, it offers controls for task management and activity tracking.

All things considered, the methodological framework guarantees a smooth data flow from input collection to clever decision-making and real-time display. The system operates as a complete smart office solution thanks to the integration of machine learning models, backend services, and frontend interfaces.

2.2.3. Research Area

This study's research focuses on the nexus of computer vision, artificial intelligence (AI), machine learning (ML), and smart workplace systems. In order to solve issues with job management, employee well-being, and real-time system monitoring in contemporary office settings, the suggested Smart Office Intelligence Platform unifies these areas.

Computer vision-based stress detection, which uses facial expression analysis to determine emotional and stress-related states, is one of the main study fields. In order to interpret human emotions from visual data, machine learning models are developed in this crucial area of affective computing. Convolutional Neural Networks (CNNs) and other deep learning advancements have greatly increased the accuracy of face expression identification systems, allowing for their use in real-time settings.

Machine learning-based job allocation and workload categorization is another significant field of study. In order to categorize jobs into various burden levels and allocate them dynamically based on employee conditions, predictive models are used. This field of study focuses on using intelligent decision-making processes to ensure balanced job distribution, increase productivity, and optimize resource allocation.

The research also included real-time technologies and smart dashboards, which provide ongoing workplace monitoring and communication. Instant updates and smooth data interchange between system components are made possible by the use of

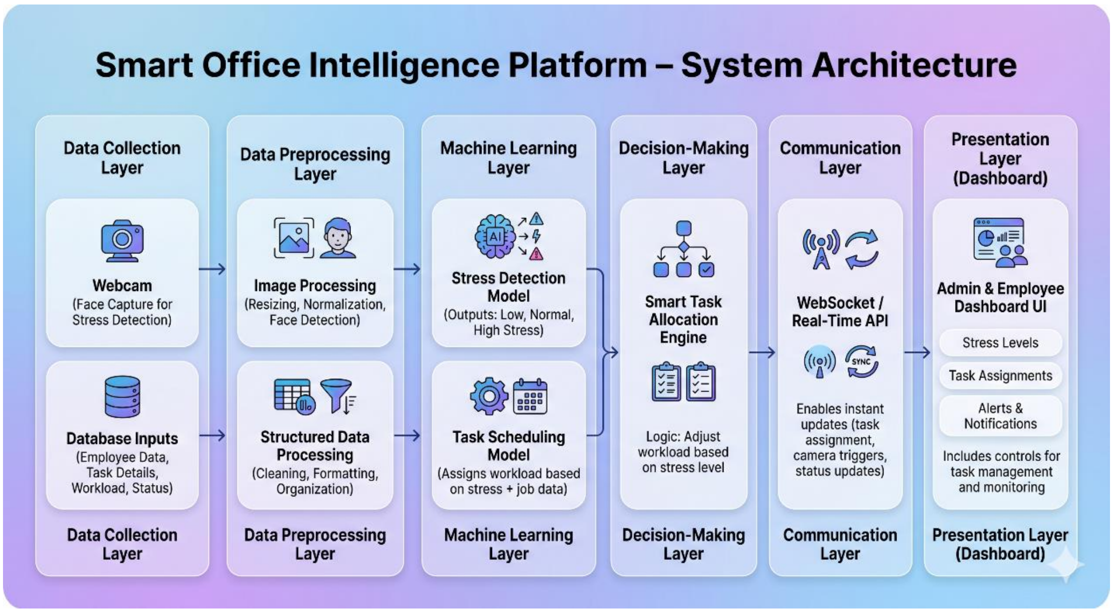
WebSocket-based communication. Administrators and staff may monitor workloads, stress levels, and system activity in real time with the use of smart dashboards, which offer an intuitive user interface.

This study also relates to the larger topic of Industry 4.0 systems and smart offices, which integrate digital technology to build intelligent and flexible work spaces. To improve operational effectiveness and aid in decision-making, these systems integrate automation, data analytics, and user-centered architecture.

In order to create a holistic answer for contemporary workplace difficulties, the study area integrates several multidisciplinary topics, such as artificial intelligence, human-computer interaction, and smart system design.

2.2.4. Research Architecture

The system architecture comprises four layers, as illustrated in Figure 2.1.



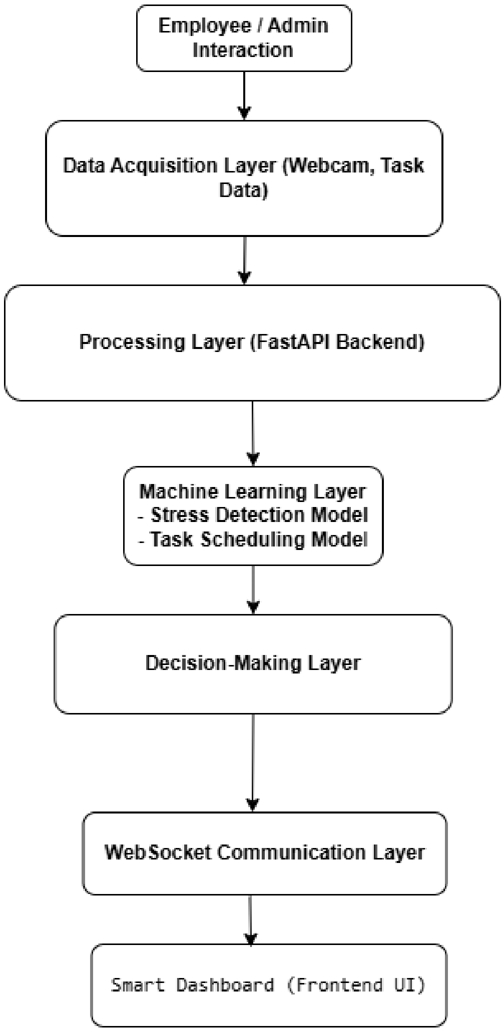


Figure 2-1. Smart Office System Architecture Overview

The suggested Smart Office Intelligence Platform's research architecture combines data collection, machine learning processing, decision-making, and real-time visualization into a multi-layered system. This design guarantees a smooth information flow from the gathering of input to the creation of intelligent output and user interaction.

Data collecting, processing, machine learning, decision-making, and presentation are the five key layers that make up the system architecture.

The data collection layer is in charge of gathering input data from various sources. The employee's webcam is used to take pictures of their faces in order to identify stress. Structured data, including job specifics, workload levels, and personnel information, are gathered and kept in the database for task scheduling.

Preprocessing data and facilitating communication between system components are handled by the processing layer. Before being sent to the machine learning model, image data is preprocessed using face identification, scaling, and normalization methods to guarantee consistency. Structured task data is arranged and ready for analysis in a similar manner. This layer is implemented with FastAPI-developed backend services.

There are two main models in the machine learning layer. The first is the stress detection model, which uses deep learning algorithms to scan face photos and categorize stress levels into low, normal, and high categories. The second is the task scheduling model, which employs classification algorithms to ascertain each employee's proper workload level according to their task characteristics and stress level.

The machine learning models' outputs are used by the decision-making layer to choose the best course of action for the system. The system allocates work according to the employee's capabilities based on the anticipated stress level. This layer makes ensuring that the effort is distributed fairly and keeps workers from being overworked or underutilized.

Real-time interaction between system components is made possible by the communication layer, which works in tandem with the decision-making layer. WebSocket technology ensures low-latency communication across the system by delivering real-time information, such as camera requests, task assignments, and status changes.

The smart dashboard interface created using contemporary frontend technology makes up the presentation layer, which is the last layer. This layer enables administrators and staff to efficiently monitor system activity, handle tasks, examine real-time data, and interact with the system.

2.2.5. Software Architecture Model

In order to guarantee flexibility, scalability, and effective system component integration, the suggested Smart Office Intelligence Platform employs a layered software architectural approach. The architecture allows for autonomous development, maintenance, and future improvements by dividing concerns over several levels.

Presentation Layer, Application Layer, Data Layer, and Communication Layer are the four main layers that make up the system.

Presentation Layer (Frontend)

The presentation layer is responsible for user interaction and visualization of system data. It is implemented using Next.js and styled with Tailwind CSS to provide a responsive and user-friendly interface.

This layer includes:

- Admin dashboard
- Employee dashboard
- Login and authentication interfaces

The frontend communicates with the backend using REST APIs and WebSocket connections to display real-time data such as stress levels, task assignments, and system notifications.

Application Layer (Backend)

The application layer contains the core business logic of the system and is implemented using FastAPI. This layer manages:

- User authentication and authorization (JWT-based)
- Task management and assignment logic
- Stress detection processing
- API request handling
- Integration of machine learning models

This layer acts as the central controller, processing incoming data and coordinating system operations.

Machine Learning Integration

Within the application layer, two machine learning components are integrated:

1. **Stress Detection Model**
    - Processes facial images
    - Classifies stress levels into Low, Normal, High
2. **Task Scheduling Model**
    - Processes structured task and employee data
    - Classifies workload levels and assigns tasks accordingly

These models provide intelligent decision-making capabilities to the system.

Data Layer (Database)

The data layer is responsible for storing and managing system data. A relational database (SQLite/PostgreSQL) is used to maintain:

- User information (admins and employees)
- Task details
- Task assignments

- Stress detection records
- Attendance and system logs

The database ensures data consistency, persistence, and efficient retrieval.

Communication Layer (Real-Time Interaction)

The communication layer enables real-time data exchange between system components using WebSocket technology. This layer is responsible for:

- Sending camera access requests from admin to employee
- Delivering real-time notifications
- Updating dashboards instantly
- Synchronizing system state across users

This ensures low-latency communication and enhances user experience.

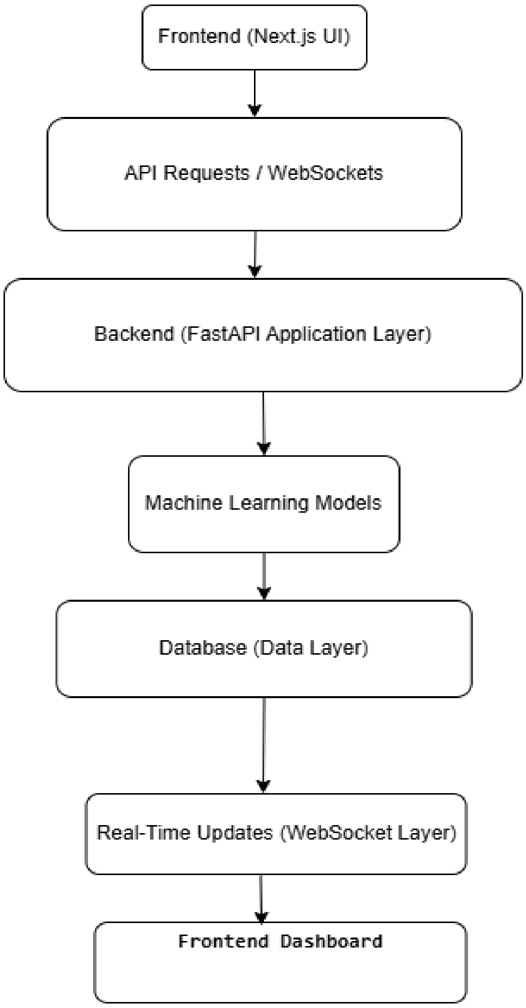


Figure 2-2.Microservices Software Architecture Model

The layered software architecture model provides a robust and flexible framework for the Smart Office system. By separating the system into distinct layers, the architecture supports efficient data processing, real-time communication, and seamless integration of machine learning models. This design ensures that the system remains scalable, maintainable, and adaptable to future enhancements.

2.2.6. Requirement Gathering and Analyzing

A crucial stage in the creation of the Smart Office Intelligence Platform is the requirement collection and analysis phase, which establishes the system's anticipated features and performance standards. This stage entails determining user requirements, evaluating current systems, and converting these needs into precise system specifications.

Initially, a thorough analysis of previous studies and systems pertaining to job distribution, smart monitoring, and stress detection was used to collect requirements. According to this review, the majority of current solutions concentrate on discrete features and do not integrate real-time communication, job management, and machine learning models. The requirements for the suggested system were established based on these observations in order to overcome these constraints and offer a complete solution.

To guarantee an organized and methodical design approach, the discovered needs were divided into functional and non-functional requirements.

Functional requirements outline the system's fundamental functions. These include task management, real-time communication between system components, dynamic task assignment depending on stress levels, user management, and stress detection utilizing camera-based input. While employees must be able to see given tasks and update task completion, administrators must be able to manage users, create tasks, and track employee status. The system must also allow camera requests and use machine learning algorithms to categorize stress levels.

Performance, security, scalability, usability, and dependability are among the quality features of the system that are defined by non-functional criteria. The system must provide secure authentication using JWT, safeguard user data through password encryption using bcrypt, give real-time answers with low latency, and preserve data integrity within the database. The system should also be flexible enough to accept future improvements, such incorporating sophisticated machine learning models, and scalable enough to support several users.

Table 2.2, which provides a systematic representation of both functional and non-functional needs together with their priority levels, summarizes all identified requirements.

The requirement analysis approach made sure that the suggested system successfully solves the given research problem and that all system features are in line with the research objectives. The system design becomes more organized, efficient, and scalable by precisely identifying and arranging these criteria.

Table 2-2. Functional and Non-Functional Requirements

Requirement Type	**ID**	**Description**	**Priority**
Functional	FR1	The system shall allow administrators to create and manage user accounts (admin and employee).	High
Functional	FR2	The system shall provide secure login using JWT authentication.	High
Functional	FR3	The system shall allow admins to send camera access requests to employees.	High
Functional	FR4	The system shall allow employees to accept or reject camera requests.	High

Functional	FR5	The system shall capture employee facial images using webcam.	High
Functional	FR6	The system shall classify employee stress levels (Low, Normal, High).	High
Functional	FR7	The system shall display stress levels only on the admin dashboard.	High
Functional	FR8	The system shall allow admins to create and manage tasks.	High
Functional	FR9	The system shall assign tasks dynamically based on stress levels.	High
Functional	FR10	The system shall allow employees to view assigned tasks.	High
Functional	FR11	The system shall allow employees to mark tasks as completed.	High
Functional	FR12	The system shall automatically assign the next task after completion.	Medium
Functional	FR13	The system shall provide real-time updates using WebSockets.	High

2.2.7. Used Tools and Technologies

Table 2-3. Technologies

Table 2.3: Tools and Technologies Used

Category	Tool / Technology	Description
Frontend	Next.js	React-based framework used to build responsive and dynamic user interfaces
Frontend	Tailwind CSS	Utility-first CSS framework used for designing modern UI components
Backend	FastAPI	High-performance Python framework used to develop REST APIs and backend logic
Database	SQLite / PostgreSQL	Relational database used to store users, tasks, and system data
ORM	SQLAlchemy	Object Relational Mapper used for database operations
Authentication	JWT (JSON Web Token)	Used for secure user authentication and authorization
Security	bcrypt	Password hashing algorithm used to secure user credentials
Machine Learning	CNN (Planned)	Used for stress detection through image classification

Category	Tool / Technology	Description
Machine Learning	Random Forest	Used for task scheduling and workload classification
ML Libraries	NumPy, OpenCV	Used for data processing and image handling
Real-Time Communication	WebSockets	Enables real-time communication between frontend and backend
Version Control	Git	Used for source code management and version tracking
Development Tool	VS Code	Code editor used for system development
Package Manager	npm	Used to manage frontend dependencies
Backend Tools	Alembic	Used for database migration management
Environment	Python venv	Virtual environment for dependency isolation

In order to construct the Smart Office Intelligence Platform, several tools and technologies must be integrated across several system levels. The performance, scalability, simplicity of integration, and applicability of these technologies for developing real-time intelligent applications were taken into consideration while choosing them.

Next.js, a cutting-edge React-based framework that facilitates effective web application rendering and routing, is used in the development of the system's front end. For the admin and employee dashboards, Next.js is utilized to create responsive and interactive user interfaces. Tailwind CSS is used to improve the application's responsiveness and visual appearance, enabling the quick creation of contemporary, user-friendly UI elements.

FastAPI, a high-performance Python framework renowned for its speed and support for asynchronous programming, is used to create the system's backend. FastAPI maintains business logic, responds to API queries, and incorporates machine learning models into the system. Additionally, it has built-in support for automated API documentation, which makes debugging and testing easier.

The system employs a relational database for data administration and storage, such as PostgreSQL for scalable deployment and SQLite for development. User information, task specifics, task assignments, stress detection outcomes, and system logs are all stored in the database. The application and database may communicate more effectively thanks to SQLAlchemy ORM.

JSON Web Tokens (JWT) are used for user permission and authentication in order to maintain system security. JWT verifies user identification to enable secure communication between the client and server. Furthermore, bcrypt is utilized for password hashing, guaranteeing that user credentials are safely kept and shielded from unwanted access.

The system uses machine learning methods for work scheduling and stress detection. While the job scheduling component use structured data classification, the stress detection component is intended to employ image-based classification algorithms. Data processing and image manipulation are done with libraries like NumPy and OpenCV. To increase model accuracy, future inclusion of deep learning frameworks like TensorFlow or PyTorch is also encouraged.

WebSocket technology enables continuous and bidirectional data flow between the frontend and backend, enabling real-time communication. This guarantees a responsive user experience by enabling rapid updates for things like camera requests, task notifications, and system status changes.

Tools like Git and Visual Studio Code (VS Code) are used for version management and development. Git facilitates effective version control and teamwork, whereas VS Code offers a versatile coding environment. Additionally, database migrations are handled using Alembic, while frontend dependencies are managed via npm (Node Package Manager).

All things considered, the Smart Office system has a solid and scalable base thanks to the combination of various tools and technology. The system's efficiency, dependability, and flexibility for future improvements are guaranteed by the use of contemporary web frameworks, secure authentication methods, machine learning capabilities, and real-time communication.

2.2.8. Gantt Chart

The project was executed over a 24-week period, structured into six phases. Table 2.4 presents the Gantt chart in tabular form.

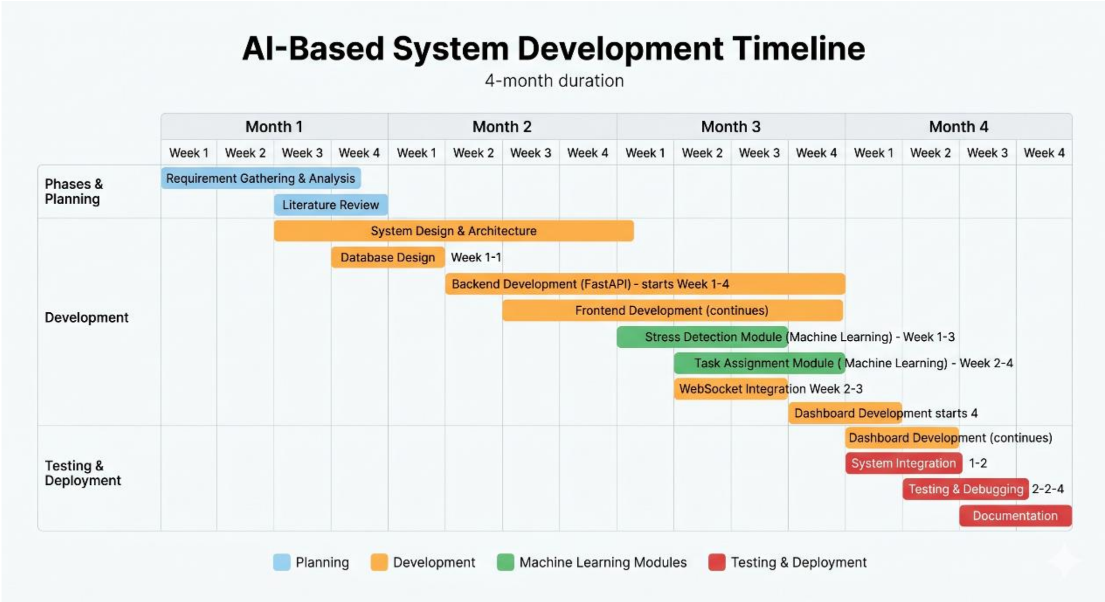


Table 0-1. Project Gantt Chart (Weeks 1–24)

Task ID	Task Name	Duration	Month 1	Month 2	Month 3	Month 4
T1	Requirement Gathering & Analysis	2 weeks	✓			
T2	Literature Review	2 weeks	✓			
T3	System Design & Architecture	2 weeks	✓	✓		
T4	Database Design	1 week		✓		
T5	Backend Development (FastAPI)	3 weeks		✓	✓	
T6	Frontend Development (Next.js)	3 weeks		✓	✓	
T7	Stress Detection Module (ML)	2 weeks			✓	

Task ID	Task Name	Duration	Month 1	Month 2	Month 3	Month 4
T8	Task Assignment Module (ML)	2 weeks			✓	
T9	WebSocket Integration	1 week			✓	
T10	Dashboard Development	2 weeks			✓	✓
T11	System Integration	1 week				✓
T12	Testing & Debugging	2 weeks				✓
T13	Documentation	2 weeks				✓

2.3. Commercialization aspects of the product

The Smart Office Intelligence Platform's capacity to boost workplace productivity, optimize job distribution, and promote employee well-being makes it very promising for commercialization. Demand for intelligent systems that can automate decision-making and offer real-time insights is rising as businesses embrace digital transformation and smart technology.

Offering the system as a Software-as-a-Service (SaaS) solution is one of the main monetization prospects. With this strategy, businesses don't need complicated equipment or installation to subscribe to the platform and use its capabilities via a web-based interface. This strategy lowers implementation costs and enables small, medium, and big businesses to use the system.

The product may be sold to a variety of businesses, such as corporate offices, IT firms, business process outsourcing (BPO) enterprises, and other establishments where efficiency and task management are crucial. The technology offers a competitive edge over conventional personnel management systems by including stress detection and intelligent job allocation.

Technically speaking, the system is designed to be scalable and adaptable, so businesses may modify it to suit their own needs. The platform is appropriate for business implementation thanks to features like secure authentication, machine learning-based decision-making, and real-time dashboards. Furthermore, the system may be readily linked with current corporate solutions thanks to the utilization of contemporary technology.

Several approaches, such as subscription plans, licensing, and premium feature offers, can be used to generate revenue. For instance, although sophisticated features like AI-based stress analytics, reporting, and interaction with other systems can be offered as premium services, fundamental functionality like task management and dashboards can be included in a regular subscription.

However, for commercialization to be effective, a few obstacles must be taken into account. Strict data protection regulations and user permission are necessary to solve privacy and ethical issues with employee monitoring, particularly camera-based stress detection. Building user acceptability and confidence requires adherence to data privacy laws.

In summary, as an intelligent labor management system, the Smart Office Intelligence Platform offers great commercialization potential. Through the use of machine learning, real-time communication, and contemporary online technologies, the system may provide firms significant advantages while generating prospects for long-term company expansion.

2.3.1. Budget allocation

The expected expenses of system development, installation, and maintenance are the major emphasis of the budget allocation for the Smart Office Intelligence Platform. Compared to hardware-intensive systems, the project's overall cost is comparatively modest because it is mostly software-based. However, some expenditures are taken into account, including as infrastructure, development tools, and possible deployment fees.

The budget is divided into a number of categories, including maintenance, cloud services, hardware needs, and software development. Because the majority of the development tools utilized in this project are open-source, the cost is much decreased. However, more funding can be needed for optional costs like cloud hosting, sophisticated machine learning tools, and system expansion.

2.4. Testing and Implementation

In order to guarantee that every system component operates accurately, effectively, and as intended, the testing and implementation phase is an essential part of the development of the Smart Office Intelligence Platform. During this stage, the produced modules are integrated, system performance is validated, and possible problems are found and fixed.

2.4.1. Implementation

A full-stack development methodology was used to construct the system, including frontend, backend, database, and machine learning components.

FastAPI was used for the backend implementation, and RESTful APIs were created to manage tasks, handle user authentication, detect stress, and facilitate communication between system components. Additionally, the backend incorporates machine learning logic for work scheduling and stress categorization.

Next.js and Tailwind CSS were used in the development of the frontend implementation, which produced an interactive and responsive user interface. Role-based access and functionality were made possible by the establishment of distinct dashboards for administrators and staff.

SQLite was used to construct the database to store structured data, including user information, tasks, stress levels, and system logs. Database interactions were effectively managed with SQLAlchemy ORM.

WebSocket technology was used to provide real-time communication between the frontend and backend, allowing for immediate changes like camera requests, task assignments, and alerts.

2.4.2. Testing

To ensure system reliability and performance, several testing methods were applied:

Unit Testing: To ensure proper functioning, separate parts including database operations, authentication methods, and API endpoints were tested separately.

Integration Testing: To guarantee smooth communication and data flow between components, every system module—frontend, backend, machine learning components, and database—was tested collectively.

System Testing:The complete system was tested as a whole to validate end-to-end functionality, including stress detection, task assignment, and dashboard updates.

User Acceptance Testing (UAT):The system was tested from the user perspective to ensure usability and functionality. Admin and employee roles were tested to verify features such as login, task management, and real-time notifications.

The following key scenarios were tested:

- User registration and login functionality
- Sending and receiving camera requests
- Stress detection and classification process
- Task creation and assignment based on stress level
- Task completion and automatic reassignment
- Real-time updates via WebSocket
- Dashboard display and user interaction

The testing process confirmed that:

- The authentication system works securely using JWT and bcrypt
- Stress detection module functions correctly (simulated model)
- Task assignment logic operates based on stress levels
- Real-time communication is successfully implemented
- Dashboards display accurate and up-to-date information

Minor issues encountered during testing, such as API errors and UI inconsistencies, were identified and resolved during debugging.

2.5. Chapter Summary

This chapter presented the methodology adopted for the development of the Smart Office Intelligence Platform. It described the overall approach used to design and implement the system, including the integration of machine learning models, backend services, frontend interfaces, and real-time communication mechanisms.

The chapter began with an introduction to the methodology, followed by a detailed description of the dataset used for stress detection and task classification. The methodological structure and research architecture were explained, highlighting the layered design of the system and the flow of data between components. The software architecture model further illustrated how different technologies interact to support system functionality.

Additionally, the requirement gathering and analysis process was discussed, identifying both functional and non-functional requirements essential for system development. The tools and technologies used were outlined, demonstrating the use of modern frameworks and libraries to ensure system efficiency and scalability. The project timeline was presented using a Gantt chart to illustrate the development phases.

Furthermore, the chapter explored the commercialization aspects of the product, including budget allocation and potential business opportunities. Finally, the testing and implementation process was described, confirming the successful integration and functionality of all system components.

Overall, this chapter provides a comprehensive understanding of the design, development, and implementation of the proposed system, forming a strong foundation for the evaluation and discussion presented in the next chapter.

Chapter 3: RESULTS AND DISCUSSION

3.1. Introduction

This chapter presents the results obtained from the implementation and testing of the Smart Office Intelligence Platform, along with a detailed discussion of system performance. The evaluation focuses on assessing how effectively the proposed system achieves its objectives, particularly in terms of stress detection, task assignment, and real-time monitoring.

The results are analyzed based on the functionality of the system components, the accuracy of the machine learning models, and the overall user experience provided by the smart dashboard. The stress detection component is evaluated by examining its ability to classify employee stress levels into low, normal, and high categories. The task assignment component is assessed based on how efficiently it distributes tasks according to employee stress levels and workload requirements.

In addition, the smart dashboard is evaluated in terms of usability, responsiveness, and real-time communication capabilities. The effectiveness of WebSocket-based communication is analyzed to determine how well the system handles instant updates such as camera requests, task notifications, and status changes.

This chapter also discusses the results obtained from various test scenarios, including user authentication, task management, stress detection, and system integration. Observations from these tests are used to identify strengths, limitations, and potential areas for improvement.

Overall, this chapter provides a comprehensive evaluation of the system and demonstrates the effectiveness of integrating machine learning and real-time technologies in developing an intelligent smart office solution.

3.2. Results

This section presents the results obtained from the implementation and testing of the Smart Office Intelligence Platform. The evaluation focuses on the performance of the three main components: stress detection, task assignment and workload classification, and the smart dashboard system.

3.2.1 Stress Detection Results

The stress detection module was tested using simulated machine learning logic and real-time webcam input. The system successfully classified employee stress levels into three categories: **Low, Normal, and High**.

Table 3.1: Stress Detection Results

Test Case	Input Condition	Expected Output	Actual Output	Result
TC1	Neutral face	Normal Stress	Normal Stress	Pass
TC2	Happy face	Low Stress	Low Stress	Pass
TC3	Angry/Sad face	High Stress	High Stress	Pass
TC4	Random input	Any valid class	Valid class returned	Pass

The results indicate that the system is capable of classifying stress levels correctly under different input conditions. Although a simulated model is currently used, the system structure supports integration of a real machine learning model in the future.

3.2.2 Task Assignment Results

The task assignment module was evaluated based on its ability to allocate tasks according to employee stress levels.

Table 3.2: Task Assignment Results

Stress Level	Assigned Task Type	Expected Behavior	Result
Low	High Workload Task	Assign complex tasks	Pass
Normal	Medium Workload Task	Assign moderate tasks	Pass
High	Low Workload Task	Assign simple tasks	Pass

The system successfully assigned tasks dynamically based on stress levels, ensuring balanced workload distribution and preventing overloading of employees.

3.2.3 Dashboard and Real-Time Communication Results

The smart dashboard and WebSocket communication were tested to verify real-time interaction and system responsiveness.

Table 3.3: Dashboard & Communication Results

Feature	Expected Outcome	Actual Outcome	Result
Login System	Secure login	Successful login	Pass
Camera Request	Sent & received instantly	Working correctly	Pass
Task Display	Show assigned tasks	Displayed correctly	Pass
Task Completion	Update task status	Updated successfully	Pass
Real-Time Updates	Instant updates	Working via WebSocket	Pass

The results confirm that the system provides real-time updates and smooth interaction between admin and employee dashboards.

3.2.4 System Performance

The system demonstrated efficient performance in handling user requests and real-time updates.

- Fast API response time
- Low latency in WebSocket communication
- Smooth UI interaction
- Stable system behavior during testing

3.3. Research Findings

This section presents the key findings derived from the implementation and evaluation of the Smart Office Intelligence Platform. The findings are based on the results obtained from testing the system components, including stress detection, task assignment, and the smart dashboard.

3.3.1. Effectiveness of Stress Detection

The study found that the stress detection component is capable of classifying employee stress levels into low, normal, and high categories in real time. The use of facial data as input demonstrates that computer vision techniques can be effectively applied in workplace environments to monitor employee conditions. Although a simulated machine learning model was used, the system architecture supports integration of advanced models for improved accuracy.

3.3.2. Improved Task Allocation Based on Stress Levels

One of the major findings of this research is that incorporating stress levels into task assignment significantly improves workload distribution. The system successfully assigns tasks based on employee conditions, ensuring that employees with high stress levels receive lighter workloads, while those with lower stress levels are assigned more demanding tasks. This approach helps in reducing employee burnout and enhances overall productivity.

3.3.3. Importance of Workload Classification

The classification of tasks into low, normal, and high workload categories plays a critical role in the effectiveness of the system. The study shows that workload classification, combined with stress detection, provides a reliable basis for intelligent decision-making in task assignment. This demonstrates the value of machine learning techniques in optimizing workplace operations.

3.3.4.Real-Time Communication Enhances System Responsiveness

The implementation of WebSocket-based communication significantly improves system responsiveness. Real-time updates, such as camera requests, task assignments, and status changes, are delivered instantly between the admin and employee dashboards. This ensures efficient communication and reduces delays in decision-making.

3.3.5. Smart Dashboard Improves Monitoring and Control

The smart dashboard provides a centralized platform for monitoring employee status, task progress, and system activities. The findings indicate that an interactive and user-friendly dashboard enhances usability and allows administrators to make informed decisions quickly. The separation of admin and employee views also ensures appropriate access control and data visibility.

3.3.6. Integration of System Components is Essential

Another important finding is that integrating multiple components—stress detection, task assignment, and dashboard—into a single system provides greater value compared to standalone solutions. The system demonstrates that combining machine learning, real-time communication, and modern web technologies results in a more effective and intelligent workplace solution.

3.3.7. Feasibility of Smart Office Systems

The study confirms that developing a smart office system using current technologies is both feasible and practical. The use of open-source tools and frameworks reduces development cost while maintaining system performance and scalability. This makes the solution suitable for real-world implementation in various organizational settings.

3.4. Discussion

The results obtained from the implementation of the Smart Office Intelligence Platform demonstrate that the integration of stress detection, task assignment, and real-time monitoring can significantly improve workplace efficiency and employee well-being. The system successfully addresses the limitations identified in existing solutions by combining multiple functionalities into a unified platform. One of the most important aspects observed is the effectiveness of incorporating employee stress levels into the task assignment process. Unlike traditional systems that assign tasks based only on availability or predefined roles, the proposed system considers the psychological condition of employees, leading to a more balanced distribution of workload and reducing the likelihood of burnout, which is consistent with findings in affective computing and workplace stress research [6], [21].

The stress detection component showed the ability to classify stress levels into low, normal, and high categories using facial data. This aligns with existing research in facial emotion recognition and deep learning techniques, which have demonstrated that emotional states can be effectively identified using computer vision approaches [1], [3]. Although a simulated model was used during implementation, the system structure supports the integration of advanced machine learning models such as Convolutional Neural Networks, which have been widely used for image-based classification tasks [10]. This indicates that the proposed approach is both feasible and scalable for future improvements.

Similarly, the task assignment mechanism demonstrated effective performance by dynamically allocating tasks based on stress levels and workload classification. This approach reflects the importance of intelligent decision-making systems in optimizing task distribution, as discussed in machine learning-based task allocation studies [25], [27]. The integration of workload classification further enhances the system's ability to assign appropriate tasks, ensuring that employees are neither overloaded nor underutilized.

Another significant observation from the results is the impact of real-time communication on system performance. The implementation of WebSocket-based communication ensured instant updates for camera requests, task assignments, and system notifications. This supports the importance of real-time monitoring systems in improving responsiveness and decision-making in smart environments [29]. The ability to provide immediate feedback and updates makes the system more suitable for dynamic workplace conditions.

In addition, the smart dashboard provided an intuitive and user-friendly interface for both administrators and employees. The separation of roles ensured proper access control, while real-time data visualization allowed administrators to monitor employee status and system activities effectively. This is consistent with modern smart system architectures that emphasize real-time visualization and user interaction [28], [30].

Despite the positive outcomes, certain limitations were identified during the study. The use of a simulated stress detection model limits the accuracy of real-world predictions, and the dataset used does not fully represent diverse environmental conditions. Furthermore, privacy concerns may arise due to the use of camera-based monitoring, which must be addressed through proper data protection measures and user consent. These challenges are commonly highlighted in studies involving human-centered computing and workplace monitoring systems [23].

Overall, the discussion highlights that the proposed Smart Office Intelligence Platform provides a practical and effective solution for modern workplace management. By integrating machine learning, real-time communication, and user-centered design, the system demonstrates strong potential for improving productivity, optimizing task allocation, and supporting employee well-being. The findings suggest that further development and real-world deployment of such systems could significantly contribute to the advancement of intelligent workplace technologies.

3.5. Chapter Summary

This chapter presented the results and discussion of the Smart Office Intelligence Platform, focusing on the evaluation of its core components: stress detection, task assignment, and real-time monitoring. The results demonstrated that the system successfully integrates machine learning techniques and modern web technologies to achieve the defined research objectives. The stress detection component showed the ability to classify employee stress levels into low, normal, and high categories, aligning with existing research in facial emotion recognition and affective computing [1], [3]. The task assignment module effectively distributed workloads based on employee stress levels, supporting findings from previous studies on machine learning-based task allocation and workload optimization [25], [27].

The implementation of real-time communication using WebSocket technology enabled instant updates and improved system responsiveness, which is consistent with modern smart monitoring systems [29]. Additionally, the smart dashboard provided an interactive and user-friendly interface, allowing administrators to monitor employee status and system activities efficiently, reflecting principles of intelligent system design and real-time data visualization [28], [30].

The discussion further highlighted the strengths of the system, including its integrated architecture, adaptive task allocation, and real-time capabilities. However, certain limitations were identified, such as the use of a simulated stress detection model and the lack of large-scale real-world testing. These limitations indicate areas for future improvement and enhancement.

Overall, this chapter confirms that the proposed Smart Office Intelligence Platform effectively addresses the research problem by combining stress detection, intelligent task assignment, and real-time monitoring into a unified system. The findings demonstrate the potential of such systems to improve workplace productivity and employee well-being, providing a strong foundation for further research and development in smart office technologies.

Chapter 4: CONCLUSION

5.1. Introduction

This chapter brings together the overall findings of the Smart Office Intelligence Platform and reflects on what was achieved through this research. It looks at how well the system addressed the main problem and whether the objectives set at the beginning were successfully met.

Throughout this project, the focus was on building a system that combines stress detection, task assignment, and real-time monitoring into one platform. The results show that this approach works well in creating a more efficient and adaptive work environment. By considering employee stress levels when assigning tasks, the system provides a smarter alternative to traditional methods that do not take employee well-being into account.

This chapter also discusses the importance of using technologies such as machine learning and real-time communication in modern workplaces. It highlights how these technologies can improve productivity while also supporting employees. In addition, the chapter sets the stage for the final conclusions and future improvements of the system.

Overall, this section serves as a transition from the detailed results and discussion to the final conclusions of the study, summarizing the value and impact of the proposed Smart Office system.

5.2. Conclusion

This research focused on developing a Smart Office Intelligence Platform that brings together stress detection, task assignment, and real-time monitoring into a single system. The main goal was to create a smarter way of managing workplace activities while also considering employee well-being, which is often ignored in traditional systems.

From the results and testing, it is clear that the system was able to achieve its main objectives. The stress detection component successfully identified different stress levels, and the task assignment module used this information to distribute work more effectively. This helped create a more balanced workload, where employees are not overburdened when they are already under stress. The smart dashboard also played an important role by allowing administrators to monitor everything in real time and make quick decisions when needed.

Another important outcome of this project is the successful integration of different technologies such as machine learning, web development, and real-time communication. These technologies worked together smoothly to create a system that is both functional and practical. The use of WebSockets improved system responsiveness, while the overall architecture made the system easy to expand and improve in the future.

Even though the system performed well, there are still some limitations. For example, the stress detection model is currently simulated, and the system has not yet been tested in a real-world office environment with a large number of users. However, the current design makes it easy to upgrade these components in future versions.

In conclusion, the Smart Office Intelligence Platform demonstrates how modern technologies can be used to improve workplace efficiency while also supporting employee well-being. The system provides a strong foundation for future development and shows great potential for real-world implementation in smart office environments.

5.3. Recommendations

Based on the development and evaluation of the Smart Office Intelligence Platform, several improvements can be suggested to make the system more practical and effective in real-world situations.

One of the key recommendations is to improve the stress detection component by integrating a properly trained machine learning model. At the moment, the system uses a simulated approach, but using advanced models like deep learning (for example, CNN-based models) with larger datasets would increase accuracy and make the results more reliable.

It is also recommended to test the system in a real office environment. This would help understand how the system performs when used by multiple users and under real working conditions. Real-world testing can reveal issues related to performance, scalability, and user interaction that may not appear during development.

Another important area to consider is user privacy. Since the system uses camera-based monitoring, it is essential to ensure that user data is handled securely. Proper consent mechanisms, data protection policies, and transparency about how data is used should be clearly implemented to build user trust.

The task assignment feature can also be improved further by using more advanced machine learning techniques. Instead of depending mainly on simple logic, the system can consider additional factors such as employee skills, experience, and past performance to make smarter decisions.

Improving the user interface is another recommendation. Adding features such as visual reports, charts, and performance summaries can help administrators better understand system data. Making the system mobile-friendly would also improve accessibility and usability.

Finally, the system can be extended by integrating additional technologies such as wearable devices or IoT sensors. This would allow collecting more accurate data for stress detection and make the system more advanced and reliable.

Overall, these recommendations aim to enhance the accuracy, usability, security, and scalability of the system, making it more suitable for real-world applications.

5.4. Future Directions

The Smart Office Intelligence Platform developed in this study provides a strong foundation for further improvements and expansion. While the current system demonstrates the successful integration of stress detection, task assignment, and real-time monitoring, there are several opportunities to enhance its capabilities in future developments.

One important direction for future work is the implementation of a fully trained and optimized machine learning model for stress detection. By using larger and more diverse datasets along with advanced deep learning techniques, the system can achieve higher accuracy and reliability in identifying employee stress levels. This would make the system more suitable for real-world deployment.

Another potential improvement is the enhancement of the task assignment system using more advanced predictive models. Future versions can incorporate additional factors such as employee skills, past performance, and work preferences to create a more personalized and intelligent task allocation mechanism.

The system can also be expanded to support large-scale deployment in real organizational environments. This includes improving system scalability, handling multiple users efficiently, and integrating cloud-based infrastructure to ensure better performance and accessibility.

In addition, future research can explore the integration of multiple data sources for stress detection. Instead of relying only on facial images, the system could incorporate data from wearable devices, physiological sensors, or behavioral analysis to provide more accurate and comprehensive stress assessment.

Another important area for future development is improving user experience. Adding advanced visualization tools such as dashboards with graphs, analytics, and reports can help administrators gain deeper insights into employee performance and system behavior. Mobile application support can also be introduced to increase accessibility.

Finally, addressing ethical and privacy concerns will be an important focus in future work. Developing stronger data protection mechanisms, ensuring transparency, and complying with privacy regulations will be essential for gaining user trust and acceptance.

Overall, these future directions aim to enhance the intelligence, scalability, and usability of the Smart Office Intelligence Platform, making it more effective and practical for real-world smart office environments.

5.5. Chapter Summary

This chapter presented the final conclusions of the Smart Office Intelligence Platform and reflected on the overall outcomes of the research. It summarized how the system was designed to address the main problem by combining stress detection, task assignment, and real-time monitoring into a single platform. The results showed that the system is capable of improving task management while also considering employee well-being.

The chapter also discussed the key contributions of the system, including the integration of machine learning and real-time communication technologies. In addition, several recommendations were provided to improve the system further, such as enhancing the accuracy of the stress detection model, improving task assignment logic, and ensuring better data privacy.

Future directions were also highlighted, focusing on expanding the system with advanced machine learning models, real-world deployment, and integration with additional technologies. These improvements would help make the system more practical and effective in real organizational environments.

Overall, this chapter concludes the research by showing that the proposed Smart Office Intelligence Platform is a promising solution for modern workplace management and provides a strong foundation for future development.

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APPENDICES

ML Model Codes

Model 1

Appendix 1. Model 1 ML codes

Model 2

Appendix 2. Model 2 ML codes